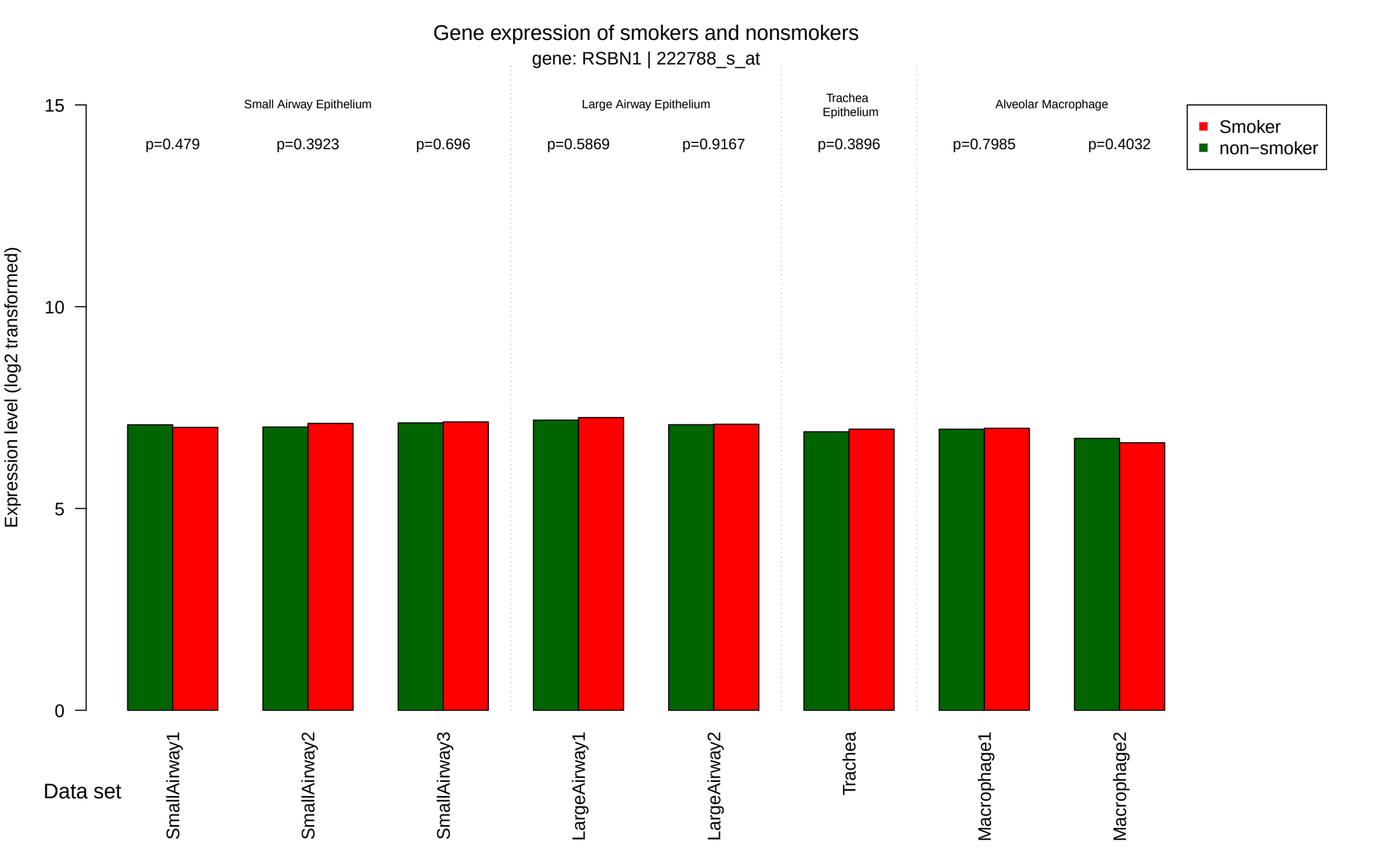
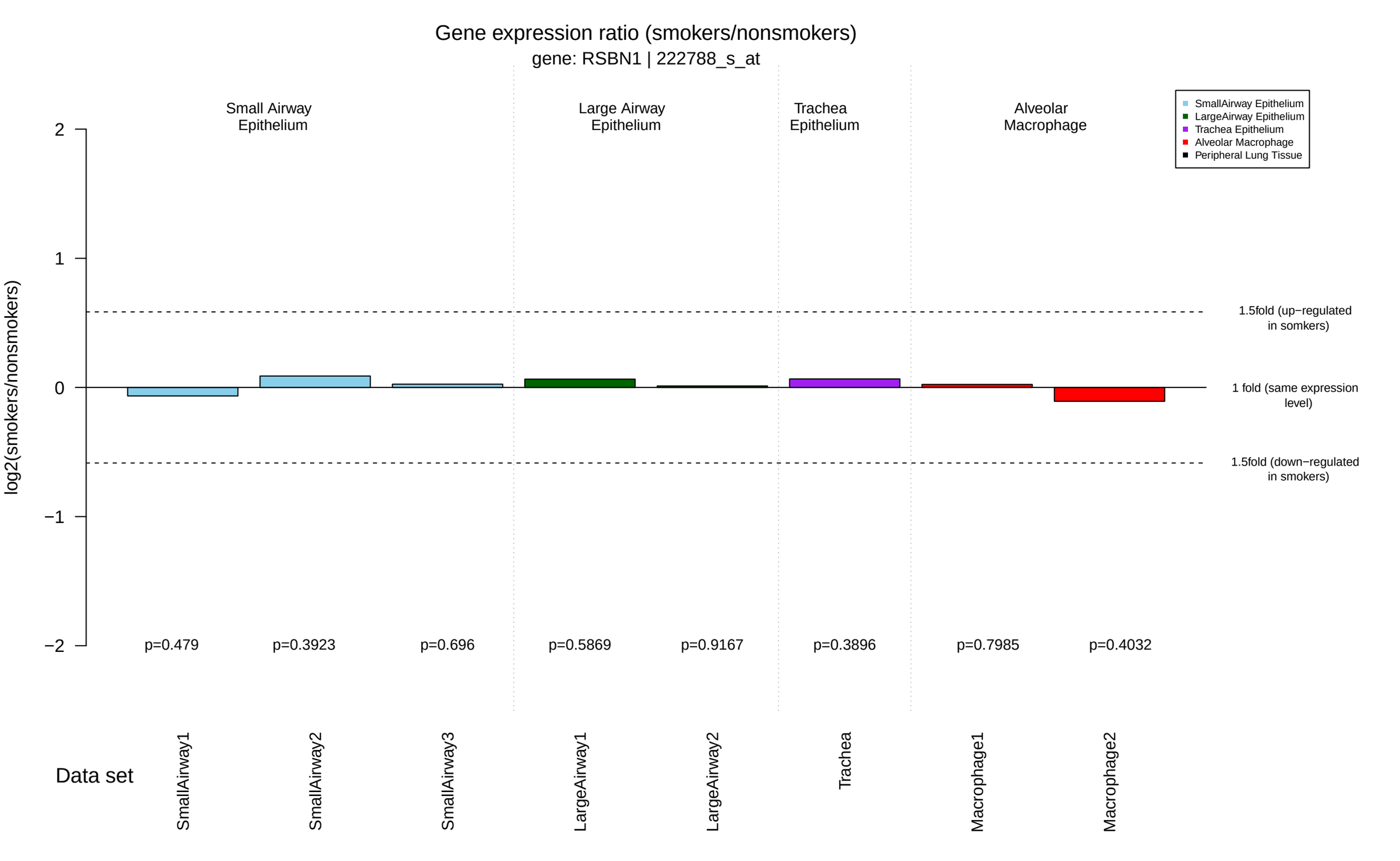
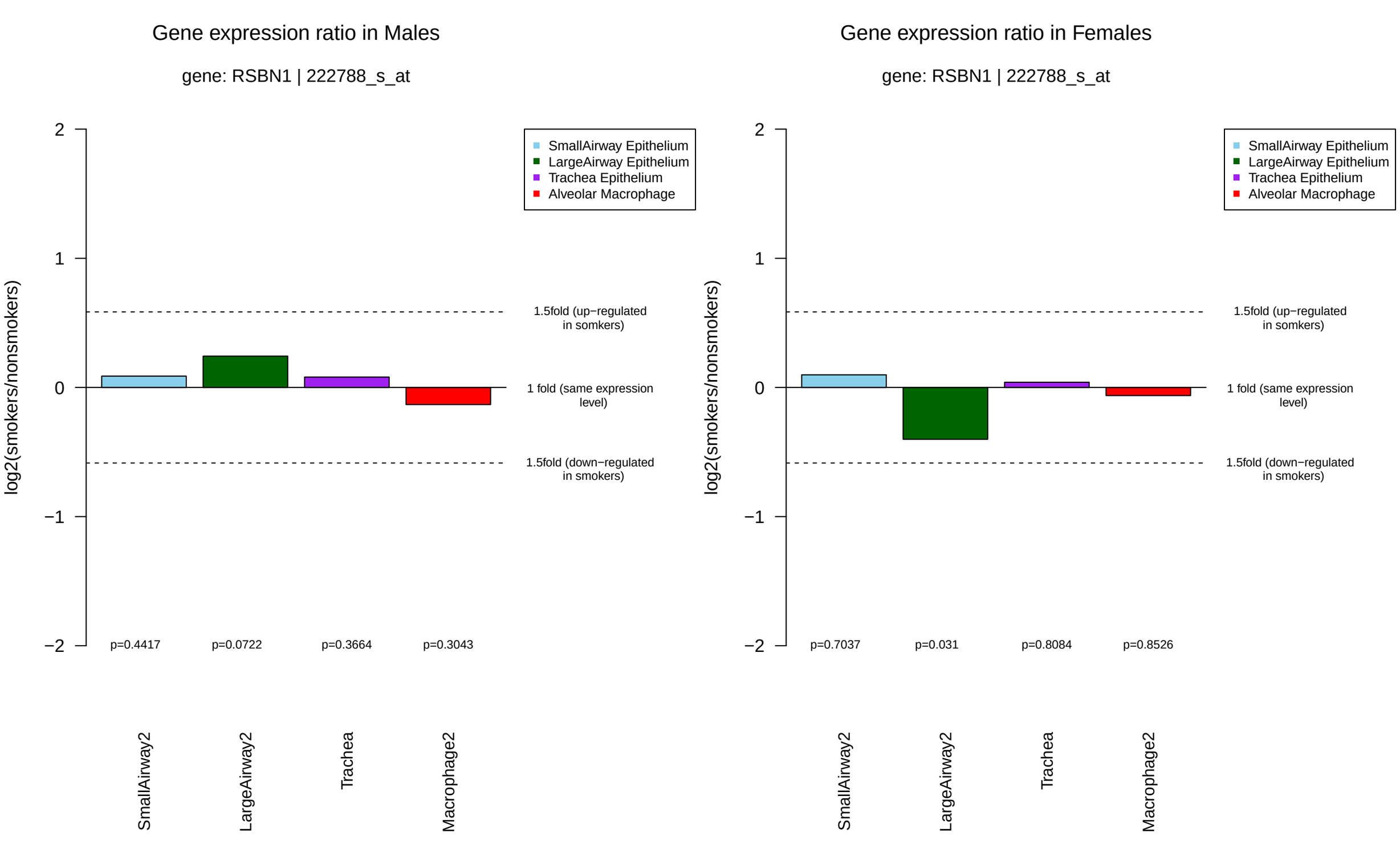
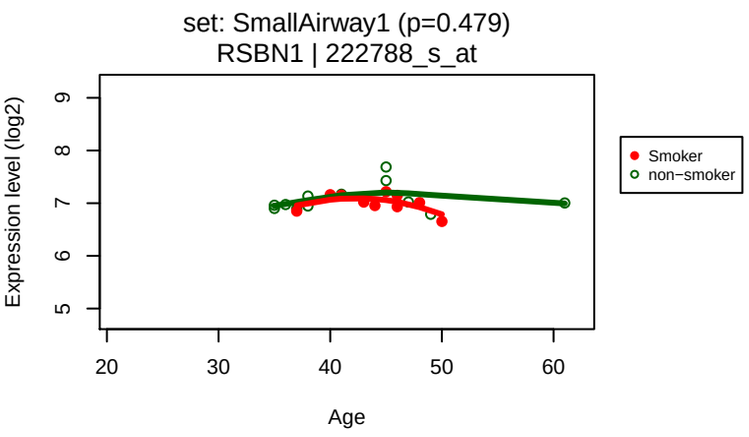
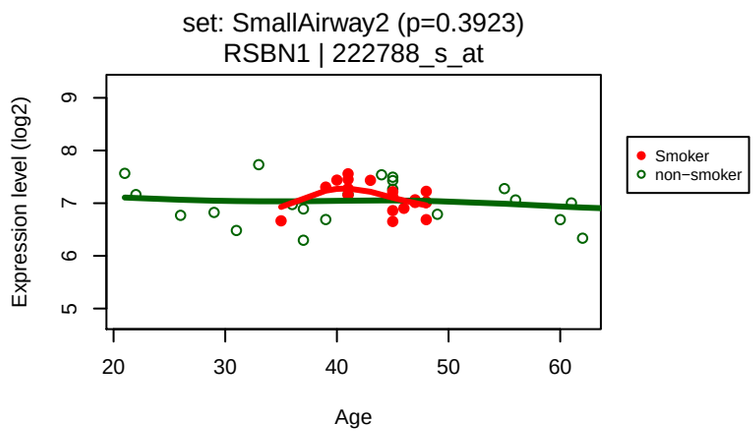
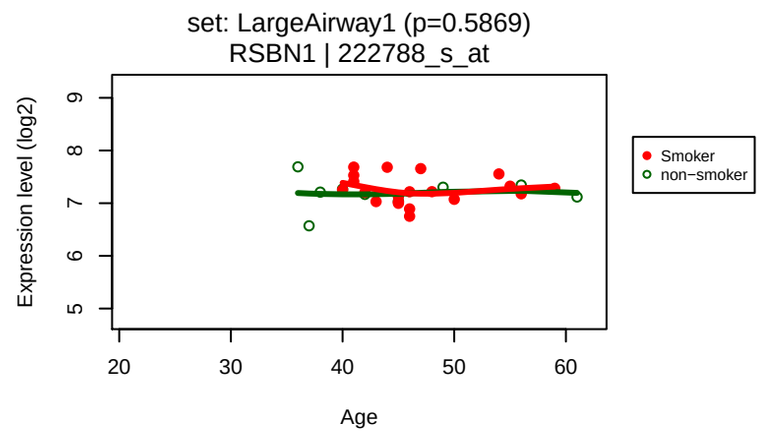

Gene expression of smokers and nonsmokers
gene: RSBN1 | 222788_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.479
p=0.3923
p=0.696
p=0.5869
p=0.9167
p=0.3896
p=0.7985
p=0.4032
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

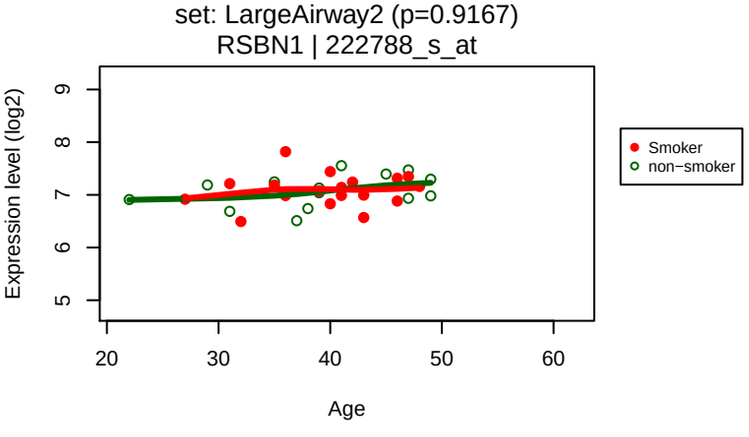

Gene expression ratio (smokers/nonsmokers)
gene: RSBN1 | 222788_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.479
p=0.3923
p=0.696
p=0.5869
p=0.9167
p=0.3896
p=0.7985
p=0.4032
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

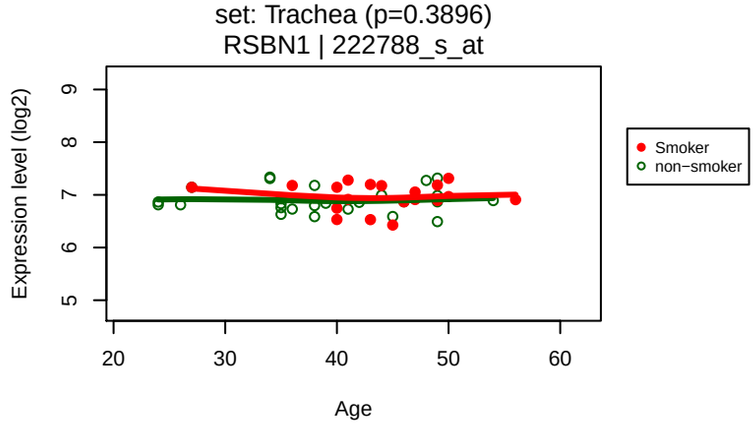

Gene expression ratio in Males
gene: RSBN1 | 222788_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4417
p=0.0722
p=0.3664
p=0.3043
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RSBN1 | 222788_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7037
p=0.031
p=0.8084
p=0.8526
SmallAirway2
LargeAirway2
Trachea
Macrophage2

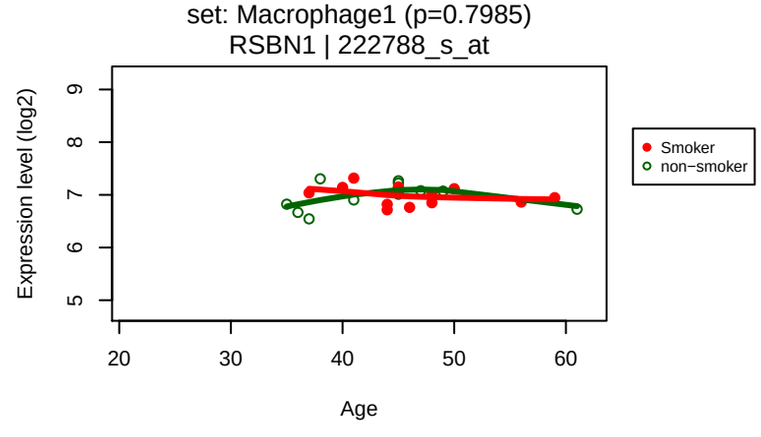

set: SmallAirway1 (p=0.479)
RSBN1 | 222788_s_at
Expression level (log2)
Age
Smoker
non−smoker

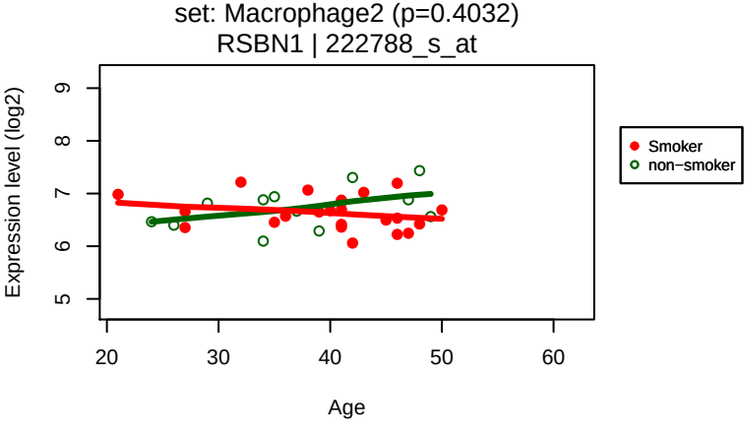

set: SmallAirway2 (p=0.3923)
RSBN1 | 222788_s_at
Expression level (log2)
Age
Smoker
non−smoker

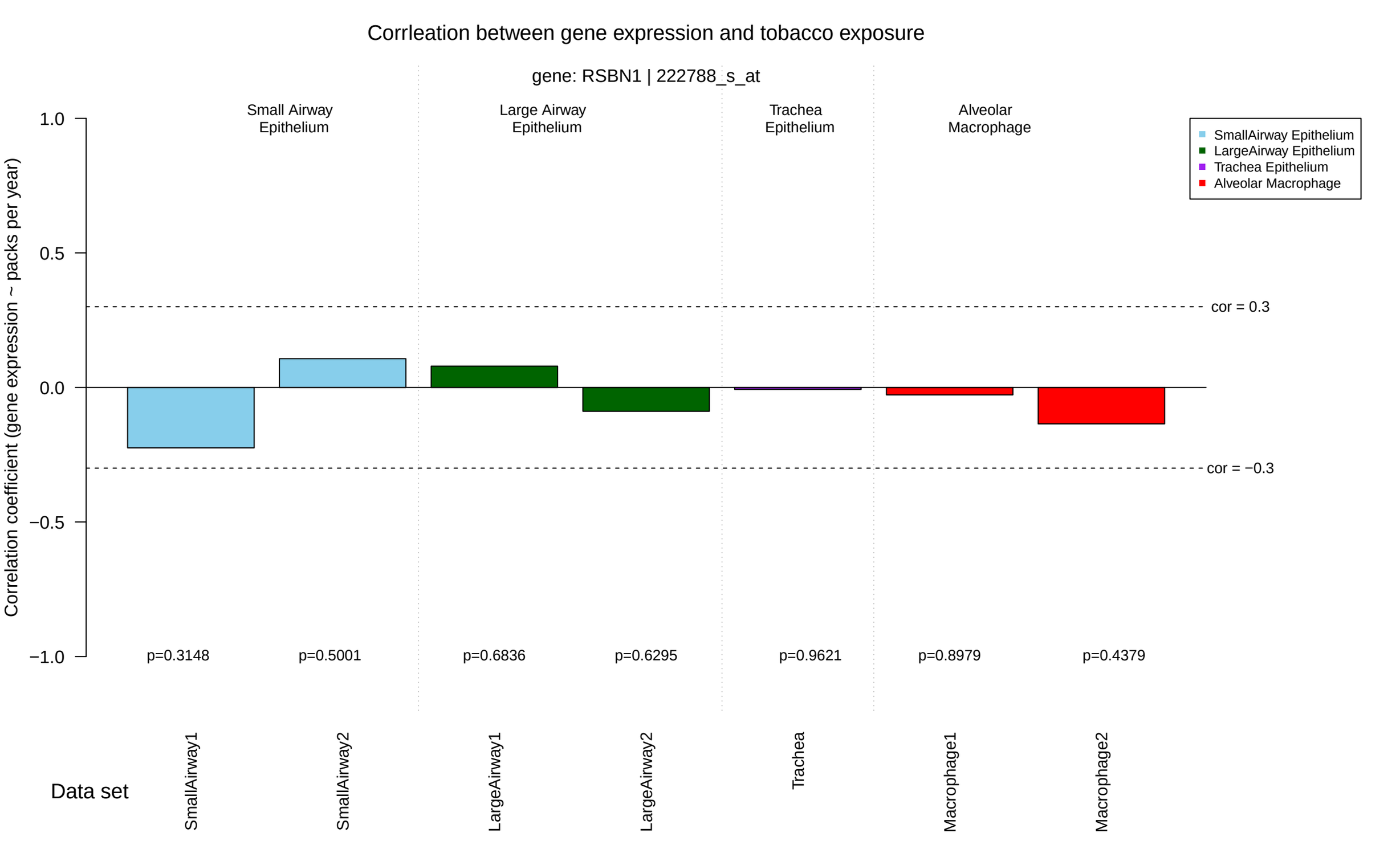

set: LargeAirway1 (p=0.5869)
RSBN1 | 222788_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9167)
RSBN1 | 222788_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3896)
RSBN1 | 222788_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7985)
RSBN1 | 222788_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4032)
RSBN1 | 222788_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RSBN1 | 222788_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3148
p=0.5001
p=0.6836
p=0.6295
p=0.9621
p=0.8979
p=0.4379
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2